

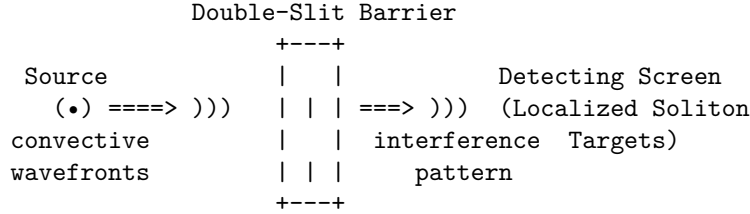
Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the double-slit experiment can be modeled from a local realist perspective for both light and electrons.

Rather than relying on wave-particle duality or wave-function collapse, this model utilizes **classical convective wavefront trains** [5.2, 10.1] and the **self-field wakes of localized current-loop solitons** [4.1] interacting with the background field-mass [6.1].

Section 15: Local Realist Modeling of the Two-Slit Experiment

We model the double-slit experiment by separating the wave dynamics of light from the self-field guiding mechanics of localized matter solitons.

15.1 The Two-Slit Experiment for Light In this system, light is modeled as a continuous train of convective electromagnetic wavefronts [10.1].



1. **Propagation and Diffraction:** A source emits a wave pulse that propagates ballistically [Section 1] according to the first-order master wave equation $DF_{\text{wave}} = 0$ [10.4]. Upon reaching a barrier with two slits, the wavefronts physically pass through both openings, generating two coherent secondary wave sources via classical Huygens-Fresnel diffraction.
2. **Superposition:** At any point $\mathbf{x}$ on the detecting screen (the receiving target), the net field bivector $F_{\text{net}}(\mathbf{x}, t)$ is the linear superposition of the two arriving wavefronts:

$$F_{\text{net}}(\mathbf{x}, t) = F_1(\mathbf{x}, t) + F_2(\mathbf{x}, t)$$

3. **Intensity Distribution:** The net local field intensity (energy density) is:

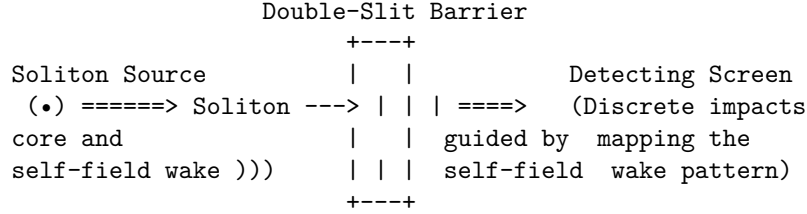
$$I(\mathbf{x}) = c^2(B_1 + B_2)^2 = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos(\Delta\varphi)$$

where the phase difference $\Delta\varphi = \frac{2\pi}{\lambda}(d_1 - d_2)$ is determined by the path length difference from the two slits to $\mathbf{x}$.

4. **Threshold-Gated Detection:** The photosensitive targets on the screen consist of localized electron solitons [4.1]. These target solitons undergo a phase-locked transition (excitation [5.2]) only if the local incident field intensity $I(\mathbf{x})$ exceeds a specific activation threshold η [10.5].

This results in a classic wave interference pattern of registered events on the screen, matching the intensity profile $I(\mathbf{x})$, without invoking any wave-particle collapse.

15.2 The Two-Slit Experiment for Electrons Unlike the purely wave-like propagation of light, the electron is modeled as a highly localized, self-bound current-loop soliton of radius r_e [4.1] carrying a mass-energy envelope.



1. **The Co-Propagating Self-Field Wake:** As the electron soliton moves at velocity u , its rotating charge-current loop [4.1] perturbs and couples with the background field-mass μ_{field} [6.1] and the Cosmic Bath [6.3], generating a real, physical, co-propagating electromagnetic self-field wake. To maintain phase-synchronization with its own induced wave wake, the spatial periodicity of this wake matches the de Broglie wavelength [5.1, 5.2]:

$$\lambda_{\text{wake}} = \frac{h}{p} = \frac{h}{m_e u}$$

2. **Interaction with the Double Slit:** When a single electron soliton is fired toward the double-slit barrier:
 - The highly localized soliton core (radius $r_e \ll$ slit width) is physically small and passes through **only one** of the two slits.
 - However, the co-propagating self-field wake is spatially wide and passes through **both** slits simultaneously.
 - The diffracted wakes exiting both slits overlap and interfere with each other, generating a real, physical interference pattern of electromagnetic field intensity $I_{\text{wake}}(\mathbf{x})$ in the space between the slits and the detecting screen.
3. **Trajectory Guidance:** As the localized electron soliton core emerges from its slit, it travels through this pre-existing, self-generated interference pattern of field intensity. The gradient of the wake's field energy density exerts a physical, classical force-gradient on the current-loop soliton:

$$\mathbf{F}_{\text{guide}} = \nabla [\boldsymbol{\mu}_e \cdot \mathbf{B}_{\text{wake}}(\mathbf{x})]$$

where $\boldsymbol{\mu}_e$ is the magnetic dipole moment of the electron loop. This force deflects the soliton's trajectory, directing it toward the constructive interference maxima of the wake field.

4. **Discrete Detection:** Upon impacting the detecting screen, the localized soliton is registered as a single, discrete event. Over many trials, the statistical distribution of these discrete impacts maps out the interference pattern, matching the classical intensity profile of the diffracted self-field wake.

This local realist model explains the double-slit interference pattern for both light and matter using classical field superposition and deterministic trajectory guidance, without requiring non-local wave-particle transitions.